Railtrack Company Guidance Note

METHODOLOGY FOR THE DEMONSTRATION OF COMPATIBILITY WITH LINESIDE EQUIPMENT

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IMPLEMENTATION

The provisions of this code of practice may be used from the date of issue.

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1 Introduction

The traction package for a modern electric traction unit will probably use IGBT or other solid state devices as switching elements to produce a variable frequency, variable voltage supply to three phase induction motors. It is possible, therefore, to generate frequencies other than 50 Hz or d.c. in the traction return current. These frequencies could cause maloperation of signalling and other equipment vital to the safety of the railway or important in railway operation.

This report describes a number of items of equipment and systems that could be affected by interference currents, in both d.c. and a.c. electrified areas. These items have been determined as part of previous studies (for example, Ref. 7) augmented by information obtained from Railtrack Zone Signalling Engineers, Racal Telecomms and other parties. Each item is examined in the light of:

- how its functionality may be affected by the train;
- what the consequences would be;
- the proposed methodology to demonstrate traction system compatibility with the lineside equipment;
- any other influencing factors.

1.1 Purpose

The purpose of this document is to provide a methodology to demonstrate compatibility with lineside equipment installed on the a.c. and d.c. electrified railway on Railtrack controlled infrastructure. This is achieved by specifying criteria/standards against which theoretical calculations or test results for new rolling stock should be compared to enable a safety case to be made.

1.2 Scope

This report is concerned with interference to fixed systems only (i.e. not trainborne) and concentrates on those that are safety critical or safety related; though systems of commercial significance are also considered. The items are arranged into broad groups according to the functions performed by them although this grouping is not precise as there are many areas of overlap. The following systems are specifically excluded as they form the subjects of separate Railtrack Company standards:

System	Railtrack Company Standard
Interlockings	RT/E/C/50013
Reed FDM Systems	RT/E/C/50015
Axle Counters	RT/E/C/50011
Train Protection and Warning System (TPWS)	RT/E/C/50012
Telecommunication systems	RT/E/C/50016

Note that this report includes only systems found on Railtrack controlled infrastructure. Equipment specific to other infrastructure is excluded.

2 Source of Documents

2.1 Railway Group And Railtrack Company Standards

1. "Code of Practice GK/RT0364 – Requirements for the Automatic Warning System", Railtrack, Issue 1, December 1997.

2. "Code of Practice GK/RT0061 – Shunter's Release, Ground Frames, Switch Panels, Gate Boxes and Shunting Frames", Railtrack, Issue 1, November 1996.

3. "Code of Practice GK/RT0190 – Emergency Voice Communications with Electrical Control Rooms", Railtrack, Issue 1, December 1995.

4. "Code of Practice for EMC between the Railway and its Neighbourhood GM/RC1500", Railtrack, Issue 1, December 1994

5. Railtrack Company Standard RT/E/PS/11762 – "Track Circuit Assister Interference Detector", Railtrack, Issue 1, December 2000.

6. Railtrack Company Standard RT/E/C/27010 – "Compatibility between Electric Trains and Electrification Systems", Issue 1, November 1997.

7. Railtrack Company Standard RT/E/C/50013 - "Methodology for the Demonstration of Compatibility with Interlockings" To be confirmed.

8. Railtrack Company Standard RT/E/C/50016 - "Methodology for the Demonstration of Compatibility with Telecommunications Systems" To be confirmed.

2.2 Previous Safety Case Documentation

9. "Juniper EE&CS safety Case: EE&CS Systems List for all Electrified Routes" AEA Technology Rail Ref. PRJ-SCSE(JUN)-RP-1, Issue 1, June 1997.

10. "Juniper Project – Electric Traction Interference: Non Track Circuit Systems" AEA Technology Rail Ref. PRJ-SCSE(JUN)-RP-16, Issue 2, September 1998.

11. "Compatibility of Class 357 Electrical Systems with Miscellaneous Systems on LTS Routes", ADtranz Ref. 3EER300000-1179, Issue D, July 2000.

12. "Class 365/5 Trains: A.C. Full Service EE&CS EMC Safety Case Assessment-Module 3 Other Signalling Systems", AEA Technology Rail Ref. PRJ-SCSE(ELAC)-SA-18, Issue 2, April 1997.

13. "Class 92 Locomotive EMC safety Assessment Volume 8-Miscellaneous Signalling Equipment", AEA Technology Rail Ref. 120-dr0442, Issue b0, May 1999.

14. "Class 92 Locomotive EMC Safety Assessment Volume 11-Miscellaneous Equipment on 750 V D.C. Railway", AEA Technology Rail Ref. 120-dr0077, Issue d0, October 1997.

15. "EE&CS Compatibility and Safety Requirements for Class 334 Trains. Hazard Analysis and Engineering Guidance", Alstom Traction, 10109.1205, Rev A, June 1999.

2.3 Other Standards

16. BS EN 50121: Railway applications. Electromagnetic compatibility, 2000

17. "Cab Secure Radio", BR Specification BR1845.

18. "VHF Band Three Radio for the National Radio Network incorporating the Train Overlay System", BR Specification 1609, Issue 2, August 1987.

2.4 Other Documents

19. "Railway Safety Principles and Guidance, Part 2, Section E: Guidance on Level Crossings", Health and Safety Executive, 1996.

20. Statutory Instrument 1992 No 2372 "Electromagnetic Compatibility", The Electromagnetic Compatibility Regulations 1992.

21. "50 Hz Single Phase A.C. Electrification: Immunisation of Signalling and Telecommunications Systems against Electrical Interference", British Railways Board, BR13422, November 1980.

3 Description of Equipment

This section gives a brief description of the items covered in this report. The items are arranged into broad groups according to the functions performed by them although this grouping is not precise, as there are many areas of overlap. Requirements for compatibility are given in section 5.

3.1 Signalling Equipment

3.1.1 Control Panels

3.1.1.1 Lever Frame with Block Bells/ Instruments

In certain areas, though rarely on electrified lines, signals and points are worked mechanically from small signal boxes by levers through wires and rodding (or in some cases, electromechanically). Interlocking of points and signals may be achieved mechanically, electrically (through conventional d.c. relay circuits) or by a combination of both. Operation of trains and communication with adjacent signal boxes uses block telegraph instruments and bells which are essentially low voltage d.c. electromagnetic devices communicating via open wire pole routes or buried cables.

3.1.1.2 Entrance-Exit (NX) Panels

An NX panel depicts the area of control on a "mimic diagram" (panel) within the control centre. The panel shows the status of routes (not set/set/occupied) using coloured lamps inset along the depiction of the line of route, the state of signals (on/off), the identity of trains (train description) and other information. The signalman sets routes by operating illuminated push buttons corresponding to the entrance and exit of the route either on the panel itself, or on a separate console. Facilities are also provided for individual operation of points, emergency replacement of automatic signals, operation of remote level crossings, release of ground frames etc.

The panel may be local to or remote from the interlocking. Communication between the panel and the interlocking may be hard wired, using conventional 50V d.c. relay circuitry or it may use multiplexing in which the status of the controls is encoded into a serial data stream using, for example, the RS 422 interface standard for transmission to the interlocking. Similarly, serial data from the interlocking is decoded into individual circuits for driving the lamps and indicators. Multiplexing is always used in Solid State Interlocking (SSI) installations and may be used in Route Relay Interlocking (RRI) installations (ref. 7). The RS 422 standard provides a degree of error protection by use of redundancy, start/stop/parity bits.

3.1.1.3 Integrated Electronic Control Centre (IECC)

The IECC is an electronic signalling control system, operated from desk-like workstations, which provides all the facilities required for the control and monitoring of a substantial area of railway.

A signaller's workstation holds four or five monitor screens providing a detailed display of the railway, train describer maps, alarms and information messages. Control is exercised either through a trackerball and push button set, or through a keyboard.

The IECC also provides:

- a train describer, with communication links to other train describer systems;
- automatic route setting;
- technician facilities for monitoring and diagnosing faults.

An IECC is composed of a number of sub-systems, depending upon the size and complexity of the scheme it serves. Each subsystem is connected to one or both of two communications networks; the Signalling Network, which handles time-critical railway control data, and the Information Network, which handles less critical information traffic. The two networks and almost all the subsystems are duplicated for availability, using an active/standby configuration. Nearly all the software is standardised, the characteristics of each scheme being stored as data.

The IECC signalling network communicates directly with one or more SSIs using multiplexed serial data. For communication with relay interlockings, a Relay Interlocking Interface subsystem converts serial data into the Westinghouse S2 protocol for communication to the interlocking via time division multiplex links and serial to parallel conversion (ref. 7).

IECC equipment is housed within the Control Centre building in the equipment room and operating floor and protected against EMI by normal industrial standards of earthing and shielding. Recent versions of equipment are tested for EMI susceptibility in accordance with the requirements of EN50121 part 4.

3.1.2 Signals

In most installations the lamp used is a type SL35, with main and auxiliary filaments each rated at 12 V, 24W but under run at 10.8 V to prolong life. Lamps are illuminated through 110 V a.c. feeds using a tail cable to transformers in the signal head. Certain types of signals use 110V a.c. 35 W long life single filament lamps as detailed below. Signals using arrays of LED light sources are currently being introduced into service (at present these are confined to Position Light Signals).

3.1.2.1 Multiple aspect (plus flashing) colour light signals

Multiple aspect signals are typically constructed with a vertical array of lamp units inside a signal head. Each unit has a double filament lamp displaying its aspect through a focused lens arrangement with coloured filter, but no reflector. Relays in the signal head effect the changeover from main to auxiliary filament and produce a filament failure alarm. Total failure of a lamp (other than the second yellow of a double yellow aspect) is detected by a relay or, in the case of SSI, in the Trackside Functional Module in the location and will restore the stop signal in rear to danger.

3.1.2.2 Position Light Junction Indicators (PLJI)

The PLJI is provided at junction signals with a speed of 40 mph or more on the fastest diverging route, to give an indication of route. This is by a line of five lunar white lamps for each diverging route. In modern installations these are SL35 but with only one filament in use. The lamps are driven by 110 V:12 V transformers in the signal head, one transformer for each diverging route and a separate transformer for the pivot lamp. A lamp proving relay, or a function in the TFM, in the location, will ensure that 3 out of 5 lamps, are alight before permitting the main signal to show a proceed aspect for a diverging route.

3.1.2.3 Position Light Signal

Position light signals are employed either to provide subsidiary aspects on main signals or as independent shunt signals, usually mounted at ground level and known as Position Light Ground Signals (PLGS). The subsidiary signal is mounted adjacent to the signal head of a main running signal. Two lunar white lights at 45° provide a ‘Shunt’ or ‘Call-on’ aspect when illuminated.

A PLGS, usually ground mounted has three lamps providing a horizontal lunar white and red aspect for stop and two lunar white lights at 45° for proceed. A recent variant provides four light sources; two horizontal red lights for stop and two white lights at 45° for proceed.

A similar signal with two horizontal red lights is used as a limit of shunt marker. An illuminated board stating “LIMIT OF SHUNT” (LOS) is also used. All position light signals use 110 V a.c. 35W long life single filament lamps or LED sources.

3.1.2.4 Banner Repeater Signal

The Banner Repeater is used for repeating the aspect of main signals where the sighting distance is inadequate. It displays a black band on a white background, horizontal for signal at danger or light out, inclined at 45° for a proceed aspect and the light on.

The aspect is obtained by extinguishing a combination of white lights supplied by optical fibre from separate "off" and "on" halogen lamps or by electromagnetic displacement of an arm against an illuminated white background with the arm proved in the signal aspect in the rear.

3.1.2.5 Multi Lamp Route Indicator(MLRI)

An MLRI will provide indication(s) of route(s) at a junction signal for speeds of 40 mph or less. A letter or figure is provided on a 7x5 or 7x7 light array, sometimes duplicated for two character displays. Lamps can be wired in parallel groups or optical fibre can be used. Where a reduction of speed of more than 10 mph relative to the fastest route is required, then illumination of sufficient lamps to provide a recognisable letter or figure is required.

3.1.2.6 Stencil Route Indicator, CD, RA and Signal OFF Indicators

Stencil route indicators provide a letter or number indication for the route(s) of a position light signal. They generally incorporate 110 V a.c. lamps behind the stencil, of the indication to be displayed. Modern indicators employ fibre optics for light transmission. Similar indicators are used for Right Away (RA), Close Doors (CD) and Signal Off indicators at station platforms. The supply voltage is 110 V a.c., 35W.

3.1.2.7 Signal Passed at Danger (SPAD) Indicator

SPAD Indicators are installed a few metres beyond a stop signal to indicate to the driver that the train has passed the stop signal at danger. A SPAD Indicator resembles a conventional 3 aspect signal but has a blue, rather than black, backboard. Normally (i.e. when indicating proceed), no lamps are illuminated. When indicating stop, the top and bottom lamps display flashing red. The centre lamp displays steady red.

3.1.3 Point Motors and Actuators

Conventional point machines are driven by series wound d.c. motors fitted with snubber circuitry (low resistance shunt) to bring the motor rapidly to rest at the end of the movement. In a.c. electrified areas, to prevent false operation due to stray traction current, a permanent magnet may be used instead of a field winding.

In a clamplock machine, the motor operates a hydraulic compressor rather than moving the switch rail directly. D.C. operated solenoid valves are used to operate the points via hydraulic pressure.

In pneumatically operated points, the motor operates a pneumatic compressor rather than moving the switch rail directly. D.C. operated solenoid valves are used to operate the points via pneumatic pressure.

3.1.4 Point Detection Circuits

Points are detected in their correct position (normal or reverse) and locked by the mechanical operation of electrical contacts. To maximise immunity to traction current interference, these contacts may be fed from supplies at various frequencies as listed below:

d.c.	non electrified lines, a.c. electrified lines
50 Hz	d.c. electrified lines
83.33Hz	dual (d.c. and a.c.) electrified lines
audio reed	d.c. and a.c. electrified lines
pseudo-random binary sequence	SSI installations

3.1.5 Line Circuits

D.C. line circuits are used in a.c. electrified and non-electrified areas. The receiving unit for d.c. line circuits is a d.c. relay. A d.c. supply is switched on or off and the relay at the receiving end will either energise or de-energise accordingly. In a.c. electrified areas the d.c. relays are immune to a.c. therefore EMI in the form of a.c. will not cause them to operate falsely. Line circuits in 25 kV 50 Hz traction areas are almost invariably fed at 50 Volt d.c., and have a contact of the controlling relays in both the outward and return conductors (double cut). If the complete circuit is required to exceed 2 km, a repeater is used at 2 km so as to limit the length of exposure of the conductors to induced interference. Relay coil resistances are dependent on type and manufacturer, but are typically about 1000 Ω, so that the line circuit power is of the order of 3 W.

The immunisation of this equipment was primarily designed to protect it from false operation by 50 Hz currents present in the traction system. This immunity will provide protection against interference from other frequencies, for example mains harmonics as a result of waveform distortion due to traction operation.

3.1.6 Automatic Warning System (AWS)

AWS is an aid to the driver which uses trackside and trainborne equipment to draw attention, by audible warnings in the cab, to the approach to certain restrictive signals, signs, level crossings and speed restrictions. The brakes are applied automatically if an AWS warning is not acknowledged within a certain time. A visual reminder is presented to the driver after acknowledgement of the warning and retained until the next set of AWS track equipment is encountered.

The track equipment consists of magnets, mounted centrally between the rails such that their magnetic fields can be detected by the AWS trainborne equipment of any train which passes over them. There are two basic forms of such magnets: a permanent magnet which has its South pole uppermost, and an electromagnet which, when energised, has its North pole uppermost. An electromagnet is always associated with a signal, and is energised only if that signal is displaying the "clear" aspect. High strength magnets are used on d.c. electrified lines in conjunction with reduced sensitivity receivers (to give greater immunity from false operation due to traction current). Permanent magnets fitted with suppressor coils may be used on bi-directional lines to prevent an AWS warning on a train travelling in opposition to the signalled direction.

The trainborne equipment consists of a bistable reed receiver which responds to the fields produced by the magnet, relay logic circuitry (electronic circuitry in the case of trains fitted with TPWS), warning horn, audible clear indicator (bell), electromechanical reminder device ("sunflower") and an interface with the braking system. A standard receiver shall respond to a flux density of 3.1 mT or greater. It shall not respond to a flux density of 1.5 mT or less. A reduced sensitivity receiver shall respond to a flux density of 5.0 mT or greater. It shall not respond to a flux density of 3.5mT or less.

In operation, the permanent magnet (South pole) initiates the warning process within the trainborne equipment. If an electromagnet (North pole) is detected within a short time delay, the equipment sounds the bell and cancels the warning process. Otherwise the horn sounds. The driver is required to acknowledge this warning, after which a visual reminder is given in the cab by the "sunflower" indicator. Failure to acknowledge within a given time, leads to an automatic application of the train brakes. Definitive information on AWS is given ref. 1

Maximum permitted levels of flux density that may be generated by rail vehicles at rail level are given in section 5.1.

3.1.7 Signalling Cables

The types of cable to be used in signalling installations are defined in the Signalling Installation Handbook. C2 type multicore cable 2, 4, 7,10, 12 or 16 cores is commonly used to supply signalling equipment at 110 V a.c.. 50 V d.c. line circuits are commonly provided by type B2 cable of section 0.75 mm^2, 1.5 mm^2 or 2.5 mm^2 with 4, 7, 10, 12, 19, 27, 37 or 48 cores. A version of 12 core cable is available containing twisted pairs.

To avoid danger and prevent damage to cables and equipment, circuit length is limited to 2 km in a.c. electrified areas such that longitudinal induced voltage does not exceed 60 V rms on full traction load or 430 V rms under fault conditions (from CCITT directive on telephone circuits reproduced in reference 20). More detailed curves giving maximum permitted induced voltage against exposure time are given in British Standard BS7354.

Tail cables commonly use C2 type 50/0.25mm multicore cable having 2, 4, 7,10, 12 or 16 cores or single core type C1. A maximum length of 200 m applies in a.c. electrified areas.

An SSI Trackside Data Link (TDL) consists of an individual screened twisted pair copper cable having a nominal impedance of 100 ohms to specification BR 1932, the equipment at each location being connected to it in parallel via Data Link Modules (DLMs). The TDL is limited in length to 10 km (where data links run parallel to a.c. electrified lines, isolating transformers are inserted at 2 km intervals).

SSI messages incorporate a number of features to reduce their susceptibility to traction interference:

- the signal levels are much higher than would be permitted over conventional shared telecommunications circuits (4.5V ± 0.25V at the transmitter); as a consequence they have a high signal-to-noise ratio and are relatively immune to interference from traction sources;

- the SSI messages are represented by complex signals which cannot conceivably be replicated by any known form of traction system;

- two levels of coding are employed and these provide a very secure means of distinguishing between data signals and noise, and of detecting messages which have been corrupted;

- the TDLs are duplicated, and provided that the interference mechanism does not affect both links, the SSI can continue functioning normally even if the data on one link is corrupted continuously;

- the occasional corruption of isolated messages on both links has no significant effect other than possibly slightly delaying the response of the SSI system to changes; such delays are completely insignificant in normal operation;

- in the unlikely event that SSI messages from both data links become continuously corrupted, the various modules of the SSI system are forced into safe (restrictive) states;

3.1.8 Ground Frame Release and Detection Systems

The controls and interlocking of ground frames is described in Group Standard GK/RT0061 [ref 2]. Release may be obtained by:

(a) the presence of a stationary train in a position where it is safe to operate the ground frame for that train.

(b) a key attached to a token or train staff or held by an authorised person.

(c) an electrical release from the controlling signal box or remote interlocking, interlocked with the relevant routes.

In the case (c), the link from interlocking to ground frame may utilise direct wire or Reed FDM. In an SSI installation, the release may be obtained via a TFM attached to an SSI datalink.

3.1.9 Emergency Signalling Power Supplies

Emergency power can be derived from:

a. Traction supplies (a.c. or d.c.). On 25 kV a.c. lines, the signalling supply points (typically 50kVA – 100kVA) are permanently connected to the 25 kV OLE through a fuse and isolator and are automatically regulated to give a closely defined output (typically 240V or 650V) over a wide range of OLE voltage. An automatic sensing circuit detects a loss of the normal 50 Hz local supply and switches the SSP emergency supply into use. Reversion to the normal supply is not automatic however, and is arranged manually at a convenient time to suit maintenance and operating needs.

b. Standby diesel generator sets. These would typically supply a control centre or remote relay room;

c. Secondary Batteries (trickle charged). These may supply a location case or an individual item such as a point machine that places an occasional heavy demand on the supply.

3.1.10 Track Circuit Assister Interference Detector (TCAID)

A common cause of wrong side track circuit failure is contamination of the interface between wheel and rail. To combat this problem, certain trains, particularly lightweight vehicles, are fitted with a Track Circuit Assister (TCA). This device induces a 165 kHz signal into the bogie/axle/wheel assembly such that a high voltage is induced across the wheel/rail interface. This voltage breaks down the contaminant film so that the track circuit is shunted.

In severe cases however, the TCA induced voltage is insufficient to break down the contaminant. The Track Circuit Assister Interference Detector (TCAID) is a trackside device which detects the 165 kHz signal from a TCA-fitted train and applies an additional shunt to the track circuit.

The TCAID functions by detecting the 165 kHz signal from the rails using a bandpass filter. The signal is applied to a threshold circuit, which prevents weak signals from adjacent tracks triggering the device, and to a delay circuit, which prevents momentary detection dropping the track circuit and momentary loss of detection removing the shunt. Finally, the signal drives a relay which short circuits the rails via a low pass filter to prevent the 165 kHz TCA signal itself from being shorted.

Some versions of TCAID include a separate 165 kHz signal input from a track mounted loop. The phase relationship between this signal and the signal from the rails determines the position of the train with respect to the device so making it directional.

TCAIDs have no direct connection to other signalling equipment and do not require an external power supply. (They have an internal battery). At present, TCAIDs are not permitted to be used on electrified lines.

3.1.11 Automatic Train Protection (ATP) Systems

Automatic train protection (ATP) is a system which extends the protection against the effect of human error, provided by the signalling system, to the driver. It supervises:

- train speed against permitted maximum train and line speeds;
- train braking against signals at danger, speed restrictions and buffer stops.

ATP provides the driver with information, warns if braking is required and, if necessary, intervenes by making a full service brake application. ATP will also make an emergency brake application if the train is detected as passing a signal at danger without authority or if detected as "rolling away". There are currently two pilot schemes operating on Railtrack infrastructure:

- On the partly a.c. electrified GW main line between Paddington and Bristol, including the Heathrow spur (ACEC TBL system);
- On the non electrified Chiltern Line between Marylebone and Aylesbury via High Wycombe (SEL SELCAB system)

Both schemes share a common specification but are implemented in technically different ways by the two manufacturers. Track mounted loops and/or beacons are used to transmit signalling information to the train. The characteristics of the track/train interface are as follows:

- GWML: 100 kHz +/-10 kHz FSK, data rate 25 kb/s (beacons), 1 kb/s (in-fill loops)
- Chiltern: 36 kHz +/- 400 Hz FSK, data rate 1.2 kb/s.

Transmitted messages contain a high level of redundancy through the use of parity bits. The trainborne processing contains redundant hardware.

The ATP trackside equipment interfaces with the existing signalling by sensing the current (at 110 V ac) fed to the signal lamps. A high degree of electrical separation of signalling and ATP circuits is provided by the use of high value series resistors mounted close to the lamp circuits.

3.1.12 European Rail Traffic Management System (ERTMS)

Beginning with the modernisation programme for the West Coast Main Line (WCML), it is envisaged that existing main line signalling, based on fixed block operation controlled by lineside signals, will gradually be replaced by ERTMS compatible systems.

ERTMS provides the capability to progressively enhance train control and protection functions though a series of implementation levels ultimately leading to transmission based cab signalling and ATP with moving block control for mixed traffic operation. Determination of train location depends upon track mounted transponders (Eurobalise) and odometry to interpolate between transponders. Eurobalise utilises 27 MHz from train to transponder, 4.5 MHz from transponder to train and provides about 1000 useable bits of data. Other track-train communication will be by GSMR digital radio. Other applications of Eurobalise are envisaged such as the control of tilt on curved track.

3.1.13 Level Crossings

3.1.13.1 Automatic Crossings

The characteristics of various types are given in Table 1, which is based upon information given in Ref. 18.

	Automatic Half Barrier (AHB)	Automatic Open Locally Monitored (AOCL)	Automatic Half Barrier Locally Monitored (ABCL)	Miniature Warning Lights (MWL)	Barrow Crossings
Initiation Point	27 s minimum (depends upon length of road crossing)	27 s minimum (depends upon length of road crossing)	As AOCL	40 s (user-worked and bridleway crossings) 20 s (footpath crossings)	Not fixed but usually at least 40s
Barriers	half road width	none	half road width	Full width barriers or gates operated by user.	None
Road Signal	1 yellow 2 red flashing Warning that lights continue to flash if 2nd train is coming. Audible alarm	1 yellow 2 red flashing Red "another train is coming" indicator (double track only). Audible alarm	As AHB but no sign warning of a second train	Miniature red/green lights	White lights illuminated when safe to cross
Rail Signal	None	Red flashing light at crossing. White flashing light displayed to train driver to indicate that red road signals are displayed.	As AOCL	none	None
Max Line Speed	160 km/h	90 km/h	90 km/h	160km/h	Not specified
Train Detection	Track circuit at strike in point Additional track circuits may be provided for bi-directional working or other functions. Treadle (possibly duplicated) to provide a precise strike-in point for initiating the timing sequence and to guard against non operation of track circuit by light weight vehicles. Additional treadle for second train detection (to keep barriers down) Treadle at strike out point to initiate re-opening.	Track circuits as for AHB, but treadles not always used.	Track circuits as for AHB, but treadles not always used.	Track circuits and (sometimes) treadles	Track circuits

Note: Treadles are generally mechanically actuated, but electronic proximity sensing types are in use in some areas.

Table 1 Various types of automatic level crossings

3.1.13.2 Manually Operated Crossings

Manually controlled barriers are monitored by the signalman or crossing keeper either by direct observation or remotely via CCTV. The barriers may be raised automatically by sequential operation of track circuits. In some cases the lowering of the barriers may also be initiated automatically by track circuit operation, but the signalman or crossing keeper is still required to confirm by observation that the crossing is clear before the signal can be cleared for a train.

3.1.13.3 Level Crossing Control

Level crossings are controlled by conventional hard-wired relay circuits initiated by track circuits and (sometimes) treadles. There are many detailed variations depending upon the type of crossing, whether on unidirectionally or bidirectionally signalled lines etc. Solid state level crossing controllers are currently under trial (Westrace, Harmon VHLC) on non-electrified lines. These are outside the scope of this document.

3.1.13.4 Level Crossing Flasher Units

These devices control the twin red flashing lights which are displayed at the road entrances when a level crossing is closed to road users. On level crossings fitted with barriers, the flashing lights also warn that the barriers are about to be closed or are already closed. The flashing is maintained until the train has passed over the crossing.

The flasher unit (type PCA1 manufactured by Westinghouse Signals) is a solid-state electronic unit. The two lamps are controlled by thyristors and flash ON-OFF alternately, but the ON periods overlap such that at least one lamp is always lit. Under failure conditions, the flashing action may cease, either as a consequence of the random failure of a component within the flasher unit, or by interference which causes spurious triggering of the thyristors. Nevertheless, the unit is designed to be "fail-safe" in the sense that, for most failures, if flashing ceases at least one lamp will remain lit.

3.1.14 Train Operated Warning System (TOWS)

Train Operated Warning Systems (TOWS) are used at selected track locations where conditions on site create particular risks for track workers. Typical locations are in tunnels, and where track curvature creates sighting problems.

The systems provide audible indications and are switched on only when track workers are present. In normal operation, and with the track clear, the system sounds a safe tone - an intermittent sound which indicates that the equipment is functioning correctly and that no warning is being given. When a train approaches the work site, the safe tone is replaced by a warning tone which indicates to the track staff that they must get clear of the appropriate track(s). The warning tone is maintained until the train passes the site. The absence of either a safe tone or a warning tone indicates a failure condition and must be treated accordingly.

The equipment controlling the tones is activated in accordance with inputs from the train detection equipment, normally the track circuits. The switching equipment is normally based on relay technology although devices based on solid state technology are being evaluated.

3.1.15 Treadles – Mechanical and Electrical

Treadles are devices used to detect the arrival of a train at specific point on the track. The device is triggered as the leading wheelset of the train reaches that point, and may or may not be re-triggered by the passage of subsequent wheelsets. Most treadles are of the SILEC type in which the flange of a passing wheel pushes down a mechanical arm thereby operating electrical contacts; mechanical damping is provided via an oil filled dashpot so that the arm returns to the raised position slowly. Detection of the passage of the first wheelset is ensured; separate detection of subsequent wheelsets may or may not occur depending on the wheel spacing and the speed of the train. The main application for treadles is as "strike-in" and "exit" devices for the automatic operation of level crossings.

A contactless wheel detector known as FREDDY was developed in the mid 1980s and is used in a number of locations, particularly on the lines of the former Eastern Region. This device is based on a commercially available proximity detector (operating at 100 Hz) driving a relay based control unit and was primarily intended as a low cost and low maintenance alternative to the SILEC treadle for the control of automatic level crossings. FREDDY can be configured in normally energised mode (for "strike-in" applications) and in a normally de-energised mode (for "exit" applications). It has also been used for "one train working" circuits in single line operation, and in axle counting systems.

3.1.16 Train Stops: Actuators and Detection

Train stops may be pneumatically operated (mainly LUL lines) or hydraulically operated (mainly Railtrack underground lines such as the Merseyrail loop). Both types have a spring return to the “set” position. Train stops are detected in their correct position (up or down) and locked by the mechanical operation of electrical contacts. To maximise immunity to traction current interference, these contacts may be fed from supplies at various frequencies as listed below:

d.c.	non electrified lines, a.c. electrified lines
50 Hz	d.c. electrified lines
83.33 Hz	dual (d.c. and a.c.) electrified lines
audio reed	d.c. and a.c. electrified lines
pseudo-random binary sequence	SSI installations

3.1.17 Train Describer Systems

The purpose of a train describer is to track the identity of a train through a control area, enabling it to be correctly routed either manually, or by Automatic Route Setting equipment (ARS), in accordance with the working timetable. The functions of a train describer are thus:

- to enable the signalman to input a train description for a train whose journey originates within the control area.
- to obtain the train description from the train describer of an adjacent control area for a train approaching from that area.
- to track the train across the control area from berth to berth in accordance with signalling and operator inputs.
- to display the train description of each train within the control area (plus trains arriving and departing to adjacent areas) on the panel or VDU display.
- to enable the signalman to cancel or modify a train description.
- to pass the train description to the train describer of an adjacent control area for a train departing to that area.

Train describers have been implemented using various technologies including electromechanical, relay systems and hard-wired electronic logic. Modern systems are invariably computer based using either dedicated equipment (generally duplicated for availability), or as a function of an IECC system (see Section 3.1.1.3). Inputs from the signalling, operator controls/keyboards and adjacent area train describers are time division multiplexed (polled) into the train describer computer.

Train describers are generally protected from the effects of electrical noise by filtering and screening of signal and power supply inputs.

3.1.18 Tripcock Testers

These are spring loaded ramps used to detect the position of the tripcock on the leading vehicle of trains fitted for train stop operation. They comprise a mechanical switch, using detection circuits appropriate to the type of electrification (as for train stops).

3.1.19 Train Running Away Alarm

TRAs (sometimes known as emergency alarms) are the last method of indication to fringe boxes in the event of a failure of the signalling system to identify that a train is entering their signalling area. Their function is comparable to the use of block bell emergency code, hence correct operation is of paramount importance. The transmission of railway signalling conditions is achieved using an analogue to digital converter enabling digital transmission systems to be used as the carrier. These units are fed from a 50 V d.c. supply.

3.2 Other Systems

3.2.1 Closed Circuit TV Systems

3.2.1.1 Level Crossing

Closed Circuit Television (CCTV) systems are used to remotely monitor unmanned barrier level crossings.

The signals from the cameras are carried to local processors, whence they are transmitted over telecommunications cables to the control centre (signal box) where the pictures are displayed on monitors. This transmission may take either of the following two forms:

a. sending the video signal directly over a high frequency co-axial cable with intermediate video amplifiers;

b. processing the video signal so that its bandwidth is reduced to the point where it can be sent over a standard digital 2 Mb/s channel. (Reduction to 64 kb/s is possible but not approved by Railtrack.)

Bandwidth reduction is achieved by the CODEC combining system which, at the transmitting end, combines the information from successive frames such that (most of the time) only differences between successive frames are transmitted. The process is reversed at the receiving end and, as far as is possible, the original picture is recovered. The recovered picture is clear and accurate only if successive frames are nearly identical; that is, if any changes in the picture are occurring slowly. Such a system is normally considered adequate for level crossing applications, where these limitations in picture quality can be offset against the lower cost of the transmission system.

Irrespective of the method of transmission, immunisation against longitudinally induced EMI is achieved by partitioning the cables into sections and inserting isolating amplifiers.

3.2.1.2 Driver Only Operation

CCTV systems for Driver Only Operation (DOO) are installed, or are currently being installed, at many stations. Each such system has multiple cameras and associated monitor banks, allowing the driver to observe the platform and the doors of the train to establish whether it is safe to close the doors and to move off.

These systems are designed such that, in the absence of a train, the screens are blank. The presence of a train is detected ultrasonically by a "Train Mass Proximity Detector" which activates the screens as a train approaches.

Cabling between the cameras and the monitors is via co-axial cable, the length of the cable being limited to approximately 100 metres in the worst case. Systems are immunised against longitudinal EMI by the use of in-line filters.

More advanced systems are being trialed. For example, “Redwing” utilises monitors installed within the cab operated by infrared signals from the platform.

3.2.1.3 Other Uses

Video distribution systems may also be used for passenger information (see Section 3.2.5) and non-operational purposes such as station security.

3.2.2 Hot Axle Box Detector (HABD)

Hot Axle Box Detectors are infrared sensitive devices which are arranged so as to measure the temperature of axle boxes on passing trains, with the aim of identifying bearings which are overheating to an extent that failure is imminent.

A typical installation covering one track uses two such heat detectors, one mounted outside each rail. Optical systems are used to limit the sensitivity of the sensors to a narrow field of view above each detector, such that axle boxes at both ends of each axle are scanned as the train passes the installation. A pair of wheel detectors is mounted alongside one rail - one on each side of the infrared sensor. These allow the infrared sensors to be activated and de-activated as the axle boxes pass, minimising the risk of misleading indications from extraneous heat sources. They also allow the axles to be counted, for the easy identification of defective axle boxes.

Alarms are generated if either the temperature of any axle box exceeds a pre-determined limit, or if the difference between the temperatures of the axle boxes at opposite ends of the same axle exceeds a predetermined limit. Axle counts and alarm conditions are transmitted via standard audio telecommunication circuits to the control centre (signal box) where the information is displayed and/or printed out.

A variety of equipment types exist, although most installations now in use are manufactured by the Servo Corporation (USA). The modern types use microprocessor technology to perform complex digital signal processing and to implement "intelligent" decision making processes.

3.2.3 Automatic Vehicle Identification (AVI)

AVI is used to identify “merry-go -round” wagons at mines and power station sites. It consists of passive wagon mounted transponders (see Section 3.2.4) and interrogators mounted in the track. The interrogators consist of 1500 mm x 400 mm loops which generate a magnetic field at 147 kHz. Loop current is about 5 A. The interrogators are inductively coupled to the transponders which selectively absorb energy such that their identity is reflected back to the interrogator; i.e., there is no separate up-link and down link as such. Data is transferred at a rate of 8000 b/s.

The system is used to update TOPS with “Merry-go-round” traffic movements and is also used for automatic billing functions. AVI is also used to identify vehicles at pantograph condition monitoring (PANCHEX) sites. For this application, transponders operating at 2.5 GHz are used.

3.2.4 Transponders

Transponders are a means of passing information (usually fixed) from track to train or train to track. A transponder system consists of two components:

- An Interrogator, which consists of an oscillator feeding RF energy to the transponder via a transmitting antenna. It also contains a receiving antenna, receiver and decoder for receiving data from the transponder.

- The Transponder, which consists of a (normally) passive sealed unit. It contains a receiving antenna tuned to the frequency transmitted by the interrogator. RF energy received is rectified and used to power the transponder circuits. The transponder, while powered up, continually transmits fixed data via a transmitting antenna, back to the interrogator.

  The interrogator is usually mounted on the train, the transponder on the track, though for AVI (see Section 3.2.3) the situation is reversed.

  The frequencies at which transponders are energised, and at which they reply, vary from one system to another. The reply frequency may be harmonically related to the energising frequency, or may be different entirely. For example, one such system uses energisation at 69 kHz with the response at 56 kb/s on a 10 MHz carrier. Future systems for railway application are likely to comply with a European directive limiting their frequency to 2.5 GHz and above.

  The information transmitted by a transponder is usually a fixed message identifying the device itself. Some systems may transmit variable information supplied by an external system such as the signalling. The message contains redundant parity bits to provide a measure of error detection (and sometimes correction). Applications of transponders include:

- AVI;
- Eurobalise (see Section 3.1.12);
- Train detection (though not on Railtrack lines).

3.2.5 Station Information Video Systems (SIVS)

Video monitors are widely used to provide arrival, departure and other information on station platforms and concourses. Monitors may be either monochrome or colour and have the following basic characteristics:

- 75 ohm unbalanced isolated inputs (isolated inputs provided to improve interference performance);
- 625 lines/frame;
- 50 frames/sec (non interlaced);
- switchable terminated and high impedance inputs.

Information is set up on a host computer connected to a video character generator by serial RS232 or RS422 twisted pair links. Monitors are fed from this via 75 Ohm coaxial cable. To overcome problems of interference, limited cable length and numbers of monitors per circuit inherent with coax distribution, serial data is sometimes sent via an omnibus circuit to individual digital-to-video converters located within, or adjacent to, each monitor.

3.2.6 Public Address System (PA)

Public address systems operate over an audio ± 3 dB bandwidth of around 50 Hz – 15 kHz. Small signal inputs from microphones are protected from interference by the use of screened twisted pair cable and local preamplifiers. Main amplifier outputs are at a high voltage (100 V rms or so) which also affords immunity to interference.

PA systems at large stations often have additional features such as:

- Electronic output switching enabling announcements to be made to particular platforms or areas;
- Ability to make different announcements simultaneously to different platforms/areas
- Delay and other techniques to reduce reverberation;
- Ability and make and replay pre-recorded announcements: Analogue voice signals are digitised as PCM and recorded on EPROM, often using some form of compression technique to reduce memory requirement;
- Transmission of announcements to remote stations (Long Line PA) using either frequency modulation to a carrier transmitted over physical line circuits or a digital PCM channel
- Microprocessor control of the above functions.

3.2.7 Station Clocks

A station clock system typically consists of a master clock which transmits d.c. pulses over a dedicated circuit to a number of slave clocks. Pulses are usually transmitted at 30 s intervals for analogue clocks, 1 s intervals for drop flap digital clocks. Other clocks may use individual 60 kHz time signal receivers. Time information is generally included in passenger information systems (see Section 3.2.5).

3.2.8 Fire Alarm Systems

Fire alarm systems in large buildings such as major stations typically consist of several radial networks of alarm pushbuttons, smoke detectors and flame detectors operated at 24 V d.c. from a trickle charged battery supply. Reversing the polarity of the supply allows the system to be tested without actually setting off the alarm.

Fire alarm systems can incorporate control functions such as automatic call out to the Fire Service or an external control centre, interface with security or access control systems, self-test or diagnostic facilities. A railway installation could include an interface to the passenger information and public address systems. Implementation of these control functions would typically be by an imbedded microprocessor system

3.2.9 Wheel Impact and Load Detectors (WILD/Wheelchex)

Strain gauges are used to measure vertical stress in the running rail caused by wheel impact and load. The strain gauges are directly attached to the rail and coupled to an adjacent signal conditioning unit by a screened cable. The conditioned signal is fed to a trackside computer by a second screened cable which can be up to 100 m long. The trackside computer incorporates a fax modem for transmitting stress readings over the ETD telephone network.

Problems have been experienced with interference from Class 91 locomotives, particularly where local earthing has not been employed. Additional input filtering has resolved these problems.

3.2.10 Business Data Systems

A number of such systems use local computers and ancillary equipment at stations and other locations which exchange data with central mainframes via the VHSDR (see ref. 8). These systems are not safety critical nor safety related (though TOPs may convey information on the carriage of dangerous goods), though their uninterrupted operation is important from an operational or commercial viewpoint. These systems include:

Operational

- TOPS freight and passenger vehicle, plus locomotive movement control;
- TRUST train movement reporting
- POIS passenger operating system
- PHIS passenger train historical record

Commercial

- CRS seat reservations
- APTIS ticket issue and accounting

- PBM parcels sales and routing to destination
- IMACS stores and inventory

3.2.11 Third Party Systems

There are many systems that could be adversely affected by EMI from solid state traction equipment. Examples of such systems where safety could be compromised include:

- defence systems
- air traffic control
- road traffic control
- medical electronics
- industrial equipment

Where such systems lie beyond the railway boundary, the railway infrastructure controller or train operator has virtually no control over them. A code of practice (Ref. 4) defines maximum levels and durations of magnetic field strength at a range of frequencies, at the railway boundary, that should minimise the risk of electromagnetic interference to external systems and ensure compliance with the European EMC Directive (Ref. 20). Ref. 16 contains more detailed information on permitted levels of electromagnetic interference that can be emitted by various items of railway equipment (rolling stock, signalling and telecommunications, power supplies etc) and details of conformance testing. The information relates mainly to induced or radiated interference. Where conducted interference is possible, case-specific testing/analysis would be required.

Third party systems may operate within the railway boundary, for example heart pacemakers, tractors for parcels and mail, and catering equipment.

3.2.12 Automatic Power Control (APC) Magnets

Permanent magnets, similar to AWS magnets, are mounted on sleeper ends to control the trainborne removal/restoration of power at neutral sections on a.c. electrified lines. Susceptibility to magnetic fields will be the same as for AWS and the same maximum levels of flux density, produced by trainborne equipment, will apply (section 5.1).

3.2.13 Traction Supervisory Pilots/SCADA Systems

These are telecommunications circuits designed to carry traction remote control signals and monitoring information from Electrical Control Rooms to the trackside control equipment (for example, busbar circuit breakers, motorised switches etc). In most cases these controls are carried over the trunk telecommunications network via 1200 baud or 2400 baud analogue modem circuits and the circuits are diversely routed via modem sharing devices. These circuits apply to both a.c. and d.c. traction control although the techniques differ across regions.

3.2.14 Panchex System

The dynamic performance of a pantograph can be determined by measuring the movement (uplift) it induces in the overhead wires. The Panchex system is designed to measure the uplift of the OHL during the passage of a train together with additional parameters such as wind speed and direction. Measurement are partially processed at the trackside, then sent via a telephone modem connection to a Remote Reporting Location (RRL). The RRL consists of a PC running specialist software to report Panchex information. An alarm will be generated if a pantograph uplift exceeds a defined limit.

3.2.15 Uninterruptible Power Supplies (UPS)

UPS are being installed to ensure continuity of the supply when the normal supply fails. It is assumed that UPS are designed to meet European Standards for immunity to EMI.

4 Infrastructure Assumptions

The methodology presented utilises the following assumptions:

- The lineside equipment has been installed to the standards applicable at the time of installation, and has been upgraded where changes in standards require that retrospective action to be taken.

- The lineside equipment as they currently exist, operates to an acceptable level of safety when used with the existing traction and rolling stock.

- The lineside equipment is correctly maintained in accordance with Group Standards and Manufacturer's recommendations where applicable.

- The lineside equipment is operating within the limits specified by the manufacturer, the design authority or by Railtrack as the infrastructure controller.

5 Susceptibility Analysis and Methodology for Demonstration of Compatibility

The following tables give the mechanisms by which the train traction equipment can interfere with each system listed in the preceding sections and the consequent effect on railway operation and safety. The following tables provide a summary the methodologies to be considered to demonstrate traction system compatibility with lineside equipment. Generally, the interference mechanisms fall into one or more of the following categories:

a. Interference at the specific operating frequency of the equipment

b. Interference at audio frequencies;

c. Interference at radio frequencies.

For equipment in category (a) it has generally been assumed that signals at frequencies outside the bandwidth of operation of the equipment will not affect its operation.

For the protection of equipment in category (c), compliance with the relevant requirements of BS EN 50121 (Ref. 16) respectively is considered necessary though not always sufficient to ensure compatibility.

5.1 Signalling Systems

Item No.	System	Interference Mechanism	Possible Result (Hazard)	Methodology for Demonstration of Compatibility	Notes
3.1.1	Control Panels	Inductive interference RFI	Failure of control and indication functions	• Contained within building. • External connections via line circuits are considered under item 3.1.5 • TDM links between control panels and interlocking are considered under in ref. 8.	• IECC installation within a control centre incorporates earthing and shielding to protect against EMI. • IECC is effectively decoupled from trackside signalling equipment by SSI.
3.1.2	Signals	Induced voltage in lamp circuit or tail cable causes false illumination	False aspect displayed	Compliance with permitted level of line current (Ref 6) and tail cable length (200m in ac electrified areas).	Circuits are normally 110 V ac. A few 12 V variants may be present.
3.1.3	Point Motors and Actuators	Induced voltage in cables	Points moving unintentionally causing damage/collision/ derailment	Compliance with permitted level of line current (Ref 6)	Induced current required for false actuation is so high as not to be credible.

Item No.	System	Interference Mechanism	Possible Result (Hazard)	Methodology for Demonstration of Compatibility	Notes
3.1.4	Point Detection	Conductive and induced voltages in cables	Signals cleared or replaced to danger unintentionally causing damage/collision/ derailment	Conductive interference should be minimised through maintenance to detect and correct earth faults. Control of induced voltages in trackside cables particularly at 50 Hz, 83 1/3Hz and reed frequencies as appropriate. Maximum permissible interference levels at the operating frequencies should be established though track circuit and reed FDM requirements are likely to be more onerous (RT/E/C/50002 and RT/E/C/50015 contain these limits).	SSI installations use pseudorandom binary sequences with high immunity to interference.
3.1.5	Line Circuits		False operation of the d.c. relay	The line circuits operate from 50 V d.c. supplies using a.c. immune relays. The length of exposed cabling is limited to 2 km or less (this is primarily for staff protection with respect to limiting any induced a.c. voltage). It is inconceivable that sufficient d.c. or low frequency voltage could be coupled into these circuits to operate the d.c. relay.	

Item No.	System	Interference Mechanism	Possible Result (Hazard)	Methodology for Demonstration of Compatibility	Notes
3.1.6	AWS	High external magnetic field at rail level causes demagnetisation of AWS permanent magnet	AWS WSF - loss of protection leading to increased likelihood of overrunning signals.	The electrical system shall not produce a flux density at rail level, between the running rails, of more than 24mT for times greater than 10ms or more than $24 \times 10^{-4}/\sqrt{t}$ (where t is in seconds) for times shorter than 10ms. This shall apply under all operating conditions including motoring and dynamic braking. Demagnetisation requires a high flux density (>20mT) for longer than 10 ms at a magnet site. Measurements should confirm that this requirement is met.	

Item No.	System	Interference Mechanism	Possible Result (Hazard)	Methodology for Demonstration of Compatibility	Notes
3.1.7	Signalling cables	Induced voltages in cables	False operation of line circuits	• Compliance with permitted level of line current (ref 6) • Avoidance of frequencies used for signalling equipment SSI data links use a spectrum centred on 10kHz. Normal messages cause the spectrum to be broadened, with significant components ranging from about 1 kHz to about 20 kHz. Because SSI links use wave shapes which are square (rather than sinusoidal), components exist at even higher frequencies but these are not important to the reception of the data. Thus, measurements of the psophometrically weighted traction current (item 3.1.9) are a useful criterion for the lower end of the SSI data link spectrum. The upper end of the spectrum overlaps the spectrum covered RFI tests for compliance with the ENV50121 Part 3-1 (ref. 16). Although these tests relate to a different coupling mechanism, they are a useful indication for low emissions in the relevant spectrum.	• On a.c. lines the length over which induction can occur from rail current is limited by booster transformers (where fitted) and by isolation in line circuits. • Line circuits on a.c. lines are designed to be a.c. immune. • Voltages on multicore cables are relatively high (several volts).

Item No.	System	Interference Mechanism	Possible Result (Hazard)	Methodology for Demonstration of Compatibility	Notes
3.1.8	Ground Frame release and detection	Induced voltages in cables	Unintentional release of ground frame	It is inconceivable that sufficient signals could induced into the short cable lengths to cause false operation of this equipment.	May use SSI or FDM

Item No.	System	Interference Mechanism	Possible Result (Hazard)	Methodology for Demonstration of Compatibility	Notes
3.1.9	Emerge-ncy signalling power supplies	• Train draws more than the permitted line current when signalling supply is drawn from traction supply. • Excessive currents at psophometrically weighted frequencies drawn from supply • OHL resonance can affect the voltage sensing of regulators used for standby signalling supplies, resulting in undervoltage when the burst of resonance starts and subsequently overvoltage as the system attempts to stabilise.	• Undervoltage causes trip and loss of supply to signalling system. • Interference and maloperation of telecommunica tions equipment running from emergency supply.	• Compliance with permitted level of line current (Ref 6) • The maximum psophometric current measured at the supply interface to the train under all conditions and at all times shall not be greater than: i. 10A averaged over any 20s period, and ii. 12.2A averaged over any 4s period, and iii. 13A instantaneous value, where instantaneous is defined as the weighted value of current over 20ms samples treated as a continuous waveform. The psophometric weighting shall be to the CCITT requirements which are illustrated in BR13422 (Ref. 21).. Line voltage ripple total harmonic distortion should be less than the specified level for the equipment.	

Item No.	System	Interference Mechanism	Possible Result (Hazard)	Methodology for Demonstration of Compatibility	Notes
3.1.10	TCAID	Significant level of line current within TCAID bandwidth (nominally 160 – 170 kHz)	Shunting of a track circuit other than that actually occupied by the train (RSF)	Line current should not produce a voltage across rails at TCAID installation point of more than 315 mV at 165 Hz (Ref. 5).	TCAIDs are not in use on electrified lines.
3.1.11	ATP	Induced currents into loops/beacons at 100 kHz (GWML) or 36 kHz (Chiltern Line).	Failure of ATP leading to increased risk of SPADs/collisions/derailments	Component of line current at 100 kHz should be less than a specified amount.	• Encoding of ATP transmissions minimises likelihood of correctly formatted false messages being generated. • Only electrified line fitted with ATP is Paddington-Heathrow.
3.1.12	ERTMS	• EUROBALISE affected by induced currents or RFI • GSM affected by RFI		See item 3.2.4 and ref. 8.	
3.1.13	Level Crossings	• Track circuit failure and/or treadle failure (see item 3.1.15) prevents or causes false actuation. • Inductive interference or RFI causing failure of flasher unit.	Irregular operation of crossing including lack of road/rail signals, barriers not descending.	RF emissions limited in accordance with ref. 16	

Item No.	System	Interference Mechanism	Possible Result (Hazard)	Methodology for Demonstration of Compatibility	Notes
3.1.14	TOWS	WSF of controlling track circuit	No warning of approaching train leads to injuries/fatalities	• Depends upon track circuit type. • Limits on lineside cable length and maximum train current.	
3.1.15	Treadles	• No conceivable electrical interference mechanism for mechanical treadles. • Inductive/conduct-ive interference causes failure of FREDDY to detect wheel	Non operation of barriers at automatic level crossing	Since operating frequency of FREDDY is 100 Hz, attention must be paid to screening to avoid inductive interference from traction current.	• Treadles are used as back up to track circuits • FREDDY is used at relatively few locations on the former Eastern Region.
3.1.16	Train Stops	Induced voltages in cables	Train stops raised or lowered unintentionally causing disruption/collis-ion	Compliance with permitted level of line current (Ref 6)	Induced current required for false actuation is so high as not to be credible.

Item No.	System	Interference Mechanism	Possible Result (Hazard)	Methodology for Demonstration of Compatibility	Notes
3.1.17	Train Describer Systems	• RF interference to control centre equipment • Interference to TDM or FDM datalinks (see ref. 8)	• Incorrect train descriptions • Incorrect routing	• RF emission in accordance with specification Ref. 15 • See ref. 8	
3.1.18	Tripcock Testers	As for train stops (item 3.1.16)	Train with defective tripcock allowed to run	See item 3.1.16	
3.1.19	Train Running Away Alarm	• Excessive line current. • Psophometrically weighted frequencies induced into cables and equipment	Failure to signal train in section leading to collision/derail-ment	Psophometrically weighted frequencies in line current to be less than those specified in item 3.1.9.	

5.2 Other Systems

Item No.	System	Interference Mechanism	Possible Result (Hazard)	Methodology for Demonstration of Compatibility	Notes
3.2.1	CCTV	Radiated EMI Induced voltage in cables	Partial or total loss of display	Radiated EMI in accordance with Ref. 15	
3.2.2	Hot Axle Box Detectors	• Heat generated by train at axle box level. • Electromagnetic interference with associated wheel detectors. • RFI affects trackside electronics • Interference with communications link to control centre	• False alarm • Hot axle box not identified correctly • Hot axle box not identified	• Thermal specification of trainborne equipment. • Compliance with permitted level of line current (Ref 6) • Radiated EMI in accordance with Ref. 15 insofar as it applies. • See ref. 8.	• Hot axle boxes are mainly confined to freight trains. • Wheel detector is designed to be immune to interference from currents in the rail.
3.2.3	AVI	Induced currents into transponders/interrog a-tors at 147 kHz	Wagon not identified or misidentified.	• 147 kHz component of line current to be less than a specified level. • Interrogator loop current is about 5 A, which is likely to be higher than any interference current.	Use of AVI for wagon identification is now very low or zero due to colliery closures.

Item No.	System	Interference Mechanism	Possible Result (Hazard)	Methodology for Demonstration of Compatibility	Notes
3.2.4	Transpo-nders	Induced current, RFI	Failure of transponder application	Line current to be within specified levels depending upon the application. RFI emissions to conform to ref. 16.	Only current UK applications are AVI and PANCHEX, see items 3.2.3 and 3.2.14.
3.2.5	Station Informat-ion Video Systems	See item 3.2.1	Could prevent the display of safety related messages e.g. fire evacuation.	Compliance with psophometric (item 3.1.9) and RFI (Ref. 16)	Systems using digital distribution likely to be less susceptible than those using long coaxial cables.
3.2.6	Public Address Systems	Psophometrically weighted induced currents at audio frequencies. Interference in cables used for long line PA (see ref. 8).	Could prevent reception of safety related messages e.g. fire evacuation	Psophometrically weighted frequencies in line current to be less than those specified in item 3.1.9.	Systems using digital distribution likely to be less susceptible than those using long cables carrying audio base-band.
3.2.7	Station Clock Systems	Pulses induced into system wiring arising, for example, through rapid changes of traction current due to third rail gapping.	Incorrect time displayed.	Compliance with specification for maximum line current (Ref 6)	Not safety related.
3.2.8	Fire Alarm Systems	RFI interferes with the control unit	False alarm/no alarm/fire alarm system does not interface with other systems.	The basic d.c. alarm system should not be susceptible to EMI.	

Item No.	System	Interference Mechanism	Possible Result (Hazard)	Methodology for Demonstration of Compatibility	Notes
3.2.9	Wheel Impact Detector	• Excessive line currents cause saturation of signal conditioning inputs. • Interference to telecomm links.	Failure to detect potential damage due to wheel impact.	Compliance with specification for maximum line current (Ref 6).	
3.2.10	Business Data Systems	• Interference with office equipment. • Interference with VHSDR cabling and equipment (see ref. 8).	Failure of systems or loss of data leading to commercial loss	RF emission in accordance with specification ref. 16.	Not safety related.
3.2.11	Third Party Systems	RFI, induced current	Equipment failure which could lead to injuries or fatalities.	Compliance with Code of Practice (Ref. 4) and more specifically with ref. 16. (see section 3.4.11)	No definitive standards exist regarding interference to heart pacemakers. However, experience has shown that levels of static flux density below 1mT are unlikely to upset these devices. Such levels should not normally be encountered in areas accessible to the public.
3.2.12	APC Magnets	See AWS (item 3.1.6)	Damage to trainborne traction equipment, OHL equipment	See item 3.1.6.	

Item No.	System	Interference Mechanism	Possible Result (Hazard)	Methodology for Demonstration of Compatibility	Notes
3.2.13	Traction Supervi-sory Pilots/ SCADA Systems	See ref. 8.	• Failure of traction supply. • Inability to isolate could lead to electrocution hazard. • Damage to equipment	See ref. 8.	Error detection can enable equipment to fail to safe state (isolated) if data is corrupted.
3.2.14	Panchex	• Interference to the Panchex Processor • Radio frequency interference to the AVI	• System failure • Vehicle with faulty pantograph not identified at PANCHEX site leading to OLE damage.	• The processor is contained inside a steel box, which itself inside aluminium or steel cabinet. Sufficient screening is therefore provided. • The operating frequency of the AVI equipment used on the Panchex system (2.4 GHz) is very high, so not likely to be susceptible to interference from traction equipment.	
3.2.15	UPS	Induced interference from traction return current.	Loss of supply to critical loads	Compliance with psophometric (item 3.1.9) and RFI limits (ref. 16).	

RAILTRACK

BRIEFING NOTE

RT/E/C/50001 to RT/E/C/50019 **Issue 1**
RT/E/G/50003; RT/E/G/00014 and RT/E/G/50016 **Issue 1**

Methodology for the demonstration of electrical compatibility between rolling stock and infrastructure

Issue Date: February 2003

Background / Synopsis

This suite of Standards (Codes of Practice and Guidance Notes) is intended to assist train-builders, TOCs, and ROSCOs in the production of Route Acceptance Safety Cases for electric trains, by describing a methodology for demonstrating compatibility with EE&CS systems on Railtrack's electrified routes.

The electrical susceptibility of the key items of infrastructure is described and characterised, and recommended methods for demonstrating compatibility between rolling stock and infrastructure are given.

Key Changes / Compliance

This is a new suite of documents. As the documents are Codes of Practice or Guidance Notes, compliance is not mandatory.

Implementation

The information contained in the documents may be used with immediate effect. The various documents in the suite will be published progressively during 2003.

Of Interest To:

Authors of Route Acceptance Safety Cases.

Name and Title of Author

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